Cohen 2021 Multiple Myeloma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity **GO**

54/73 samples (≥94 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 80 cells / 3 specks removed; iLISI median 1.00→14.2 of 54). Low-θ density lobes were patient batch, resolved at θ=10; one lobe carries the trajectory.

Gate 2 — module geography (non-branching) **GO**

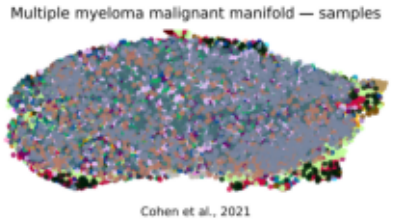
No Gast/pSom divergence into competing arms (spatial r=+0.07 — no bifurcation). A single dominant contrast opposes Crest to both (Crest–Gast r=−0.69, Crest–pSom r=−0.61), Crest rising to a terminal readout (crest-high decile mean pt=0.55). pSom>Gast, Crest≈0 globally.

Gate 3 — hub mutual exclusivity **NA**

Single continuous axis — no bifurcation and no Gast/pSom convergence hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

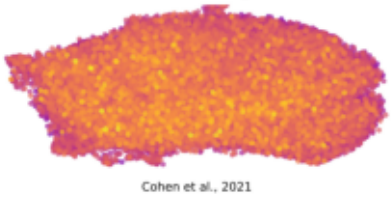
Gate 1: integration



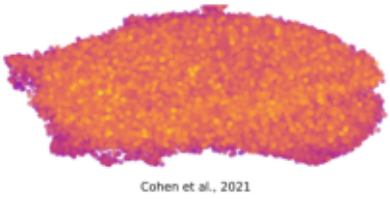
Trajectory / root



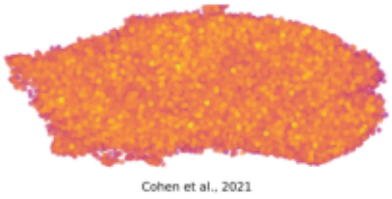
Gast



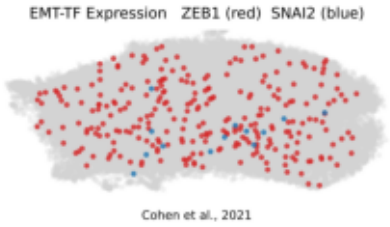
pSom



Crest



EMT-TFs



Dataset. Multiple Myeloma (Cohen et al., 2021), MARS-seq — 74 samples / 39,157 cells (master scope); malignant only: 33,599 load, 33,147 across 54 samples (≥94) after thresholding, 33,067 after island removal enter the trajectory; whole manifold (no scoping).

Verdict. ✓ Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Cohen's MM plasma-cell manifold is a single continuous gastrulation/somitogenesis → terminal-neural-crest gradient, not Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. Modules are globally weak (Gast −0.022, pSom +0.014, Crest +0.001), consistent with shallow-MARS-seq plasma cells; the axis is orthogonal to (near-absent) proliferation. EMT-TF program essentially absent — only ZEB1 detectable (~3%); Snail/Twist/Foxc/Prrx at the detection floor.

Parameter choices

Min cells/sample	≥94 (54/73; gap 117→77)	**Integration**	Harmony θ=10; PCA n_pcs=21; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 0 = Gast/pSom-high, Crest-low	**EMT-TFs**	ZEB1 (~3%); others <0.3% (absent)